

ISyE6669 Homework Week 6

Fall 2025

1. Consider the following nonlinear optimization problem:

$$\begin{array}{ll}\min & |2x + 3y| + |2x - 3y| \\ \text{s.t.} & \max\{y, x + y, x - y\} \leq 5.\end{array}$$

Reformulate it as a linear program.

Solution: Problem can be reformulated as

$$\begin{array}{ll}\min & z_1 + z_2 \\ \text{s.t.} & -z_1 \leq 2x + 3y \leq z_1 \\ & -z_2 \leq 2x - 3y \leq z_2 \\ & y \leq 5 \\ & x + y \leq 5 \\ & x - y \leq 5\end{array}$$

2. Consider the following sets. For each set, determine whether it can be represented using linear inequalities. If it can, provide a formulation using linear inequalities.

- $\{x | \max\{3x + 5, 7x - 3\} \geq 6\}$
- $\{x | \max\{3x + 5, 7x - 3\} \leq 6\}$
- $\{x | |5x - 3| \leq 4\}$
- $\{x | |5x - 3| \geq 4\}$

Solution:

- In general set defined as:

$$\{x | \max\{g_1(x), g_2(x)\} \geq c\}$$

where $g_1(x)$ and $g_2(x)$ are linear functions and c is constant, does not have to be convex and when it is not convex it can't be written using linear constraints.

However, this particular set $\{x | \max\{3x + 5, 7x - 3\} \geq 6\}$ is convex and thus can be written using linear constraints. To rewrite it using linear constraints first we can solve:

$$3x + 5 = 7x - 3 \quad \Rightarrow x = 2$$

Now we have:

$$\begin{aligned} \{x | \max\{3x + 5, 7x - 3\} \geq 6\} &= \{x | x \leq 2, 3x + 5 \geq 6\} \cup \{x | x \geq 2, 7x - 3 \geq 6\} = \\ &= \{x | x \leq 2, x \geq \frac{1}{3}\} \cup \{x | x \geq 2, x \geq \frac{9}{7}\} = \\ &= \{x | x \leq 2, x \geq \frac{1}{3}\} \cup \{x | x \geq 2\} = \\ &= \{x | x \geq \frac{1}{3}\} \end{aligned}$$

- Set $\{x | \max\{3x + 5, 7x - 3\} \leq 6\}$ is convex because it is α -level set of a convex function and hence can be rewritten using linear inequalities.

Similarly, to the previous problem we can obtain linear representation in the following way:

$$\begin{aligned} \{x | \max\{3x + 5, 7x - 3\} \leq 6\} &= \{x | x \leq 2, 3x + 5 \leq 6\} \cup \{x | x \geq 2, 7x - 3 \leq 6\} = \\ &= \{x | x \leq 2, x \leq \frac{1}{3}\} \cup \{x | x \geq 2, x \leq \frac{9}{7}\} = \\ &= \{x | x \leq \frac{1}{3}\} \cup \emptyset = \\ &= \{x | x \leq \frac{1}{3}\} \end{aligned}$$

Alternatively, as feasible set this convex set can be reformulated by introducing another variable as:

$$\{(x, z) | x \leq 6, 3x + 5 \leq z, 7x - 3 \leq z\}$$

- Set $\{x | |5x - 3| \leq 4\}$ is convex and thus can be written using linear inequalities as:

$$\{x | |5x - 3| \leq 4\} = \{x | -4 \leq 5x - 3 \leq 4\}.$$

Alternatively, we can reformulate this set by introducing new variable as:

$$\{(x, z) | z \leq 4, 5x - 3 \leq z, 3 - 5x \leq z\}$$

- Set $\{x | |5x - 3| \geq 4\}$ is not convex and thus can't be written using linear inequalities.

3. A response variable y is known to depend on a predictor x . A set of n data pairs $\{(x_i, y_i)\}_{i=1}^n$ has been collected. Formulate an optimization model for fitting the “best” straight line

$$y = a + bx$$

to the data, where best is measured with respect to the sum of absolute deviations between observed and predicted values. Can this problem be reformulated as a linear program?

Solution: This problem can be formulated as

The optimization problem can be directly formulated as

$$\min_{\forall a, b \in \mathbb{R}} \sum_{i=1}^n |y_i - a - bx_i|$$

Objective function of this problem is convex piecewise linear function. Problem can be reformulated as LP in the following way:

$$\begin{aligned} \min \quad & \sum_{i=1}^n z_i \\ \text{s.t.} \quad & y_i - a - bx_i \leq z_i, \quad \forall i = 1, \dots, n, \\ & a + bx_i - y_i \leq z_i, \quad \forall i = 1, \dots, n, \end{aligned}$$

4. Hill-O-Beans Coffee Company blends four component beans into three final blends of coffee: One is sold to luxury hotels, another to restaurants and the third to supermarkets for store-label brands. The company has four reliable bean supplies: Argentine Abundo, Peruvian Colmado, Brazilian Maximo and Chilean Saboro. The table below summarizes the very precise recipes for the final coffee blends, the cost and availability information for the four components and the wholesale price per pound of the final blends. The percentages indicate the fraction of each component to be used in each blend.

Component	Hotel	Restaurant	Market	Cost/Pound	Weekly Availability
Abundo	20%	35%	10%	0.60	40,000
Colmado	40%	15%	35%	0.80	25,000
Maximo	15%	20%	40%	0.55	20,000
Saboro	25%	30%	15%	0.70	45,000
Wholesale price/pound	1.25	1.50	1.40		

The processor’s plant can handle no more than 100,000 pounds. There is no problem in selling the final blends although the marketing department requires minimum production levels of 10,000, 25,000 and 30,000 pounds respectively for the hotel, restaurant and market blends.

Write a linear program to maximize the weekly profit of Hill-O-Beans. Make sure to explicitly define all decision variables. Implement your model in CVXPY and print out the optimal objective value. What is the optimal amount of each bland which should be produced?

Solution:

Decision variables:

- x_H : amount, in pounds, of hotel bland produced
- x_R : amount, in pounds, of restaurant bland produced
- x_M : amount, in pounds, of market bland produced

Using these decision variables and values from the table above we can express following quantities:

Coffee \ Amount	Hotel	Restaurant	Market	Total
Abundo	$0.2x_H$	$0.35x_R$	$0.1x_M$	$0.2x_H + 0.35x_R + 0.1x_M$
Colmado	$0.4x_H$	$0.15x_R$	$0.35x_M$	$0.4x_H + 0.15x_R + 0.35x_M$
Maximo	$0.15x_H$	$0.2x_R$	$0.4x_M$	$0.15x_H + 0.2x_R + 0.4x_M$
Saboro	$0.25x_H$	$0.3x_R$	$0.15x_M$	$0.25x_H + 0.3x_R + 0.15x_M$
Revenue	$1.25x_H$	$1.5x_R$	$1.4x_M$	$1.25x_H + 1.5x_R + 1.4x_M$
Cost	$0.6975x_H$	$0.65x_R$	$0.665x_M$	$0.6975x_H + 0.65x_R + 0.665x_M$
Profit	$0.5525x_H$	$0.85x_R$	$0.735x_M$	$0.5525x_H + 0.85x_R + 0.735x_M$

From the above, we get:

Objective function:

Profit: $0.5525x_H + 0.85x_R + 0.735x_M$

Constraints:

- Abundo availability: $0.2x_H + 0.35x_R + 0.1x_M \leq 40,000$
- Colmado availability: $0.4x_H + 0.15x_R + 0.35x_M \leq 25,000$
- Maximo availability: $0.15x_H + 0.2x_R + 0.4x_M \leq 20,000$
- Saboro availability: $0.25x_H + 0.3x_R + 0.15x_M \leq 45,000$
- Minimum production of hotel blend: $x_H \geq 10,000$
- Minimum production of restaurant blend: $x_R \geq 25,000$
- Minimum production of market blend: $x_M \geq 30,000$
- Maximum capacity of the processor's plant: $x_H + x_R + x_M \leq 100,000$

By putting decision variables, objective and constraints together we get the following formulation:

$$\begin{aligned}
\max \quad & 0.5525x_H + 0.85x_R + 0.735x_M \\
\text{s.t.} \quad & 0.2x_H + 0.35x_R + 0.1x_M \leq 40,000 \\
& 0.4x_H + 0.15x_R + 0.35x_M \leq 25,000 \\
& 0.15x_H + 0.2x_R + 0.4x_M \leq 20,000 \\
& 0.25x_H + 0.3x_R + 0.15x_M \leq 45,000 \\
& x_H \geq 10,000 \\
& x_R \geq 25,000 \\
& x_M \geq 30,000 \\
& x_H + x_R + x_M \leq 100,000
\end{aligned}$$

Note: There are multiple correct ways to formulate LP. Any correct **linear formulation** which resulted in correct solution, should be accepted.

By implementing the above in CVXPY we get the following solution:

Optimal value: \$55,200,
Hotel blend produced: 10,000 *lb*,
Restaurant blend produced: 32,500 *lb*,
Market blend produced: 30,000 *lb*.

5. A farm grows two types of mangoes. This year the yield of mangoes of type 1 is 1000 units and that of type 2 is 700 units. The farm sells all of its mangoes. The mangoes can be sold to a retail company and can also be sold in a local market directly. The contract with the retail company has the following details:
- (a) At least 60% of the total amount of mangoes sold to the retail company must be of type 2.
 - (b) The retail company has a demand of 1000 units of mangoes and pays \$3 per unit for both types of mangoes.
 - (c) It is possible to sell less than the demand to the retail company: If the total amount of mangoes sold to the retail company is less than 1000 units, then there is a penalty of \$0.05 per unit shortfall from 1000 units. (For example, if the farm decides to sell 900 units of mangoes to the retail company, then the farm earns $900 * 3 - (1000 - 900) * 0.05$)
 - (d) It is possible to sell more than the demand to the retail company. However if the number of units of mangoes is more than 1200, then each unit over the first 1200 units is purchased by the retail store at the rate of \$2.5 per unit. (For example, if the farm decides to sell 1300 units of mangoes to the retail company, then the farm earns $1200 * 3 + (1300 - 1200) * 2.5$)

Mangoes of type 1 sell at \$2.2 per unit in the local market and mangoes of type 2 sell for \$3.1 per unit in the local market. Any amount of mangoes can be sold in the local market. Formulate a linear program to maximize the net revenue of the farm.

Solution:

Decision variables:

- r_1 : amount of type 1 mangoes sold to a retail company
- r_2 : amount of type 2 mangoes sold to a retail company
- m_1 : amount of type 1 mangoes sold in a local market
- m_2 : amount of type 2 mangoes sold in a local market

Objective function:

The revenue of the farm is equal to the revenue the farm earns by selling mangoes to a retail company plus the revenue the farm earns from selling mangoes directly in the market, i.e.,

$$f(r_1, r_2, m_1, m_2) = f_R(r_1, r_2) + f_M(m_1, m_2)$$

Revenue the farm earns by selling mangoes directly in the market we can calculate as:

$$f_M(m_1, m_2) = 2.2m_1 + 2.1m_2$$

which is a linear function (thus both concave and convex).

To calculate the revenue the farm earns by selling mangoes to a retail company, we need to consider different cases. Since retail company pays same amount for both types of mangoes, to simplify the expressions, let's denote $x_R = r_1 + r_2$. Now we have:

$$\begin{aligned} f_R(x_R) &= \begin{cases} 3x_R - 0.05(1000 - x_R) & x_R \leq 1000 \\ 3x_R & 1000 \leq x_R \leq 1200 \\ 3600 + 2.5(x_R - 1200) & 1200 \leq x_R \end{cases} \\ &= \begin{cases} 3.05x_R - 50 & x_R \leq 1000 \\ 3x_R & 1000 \leq x_R \leq 1200 \\ 2.5x_R + 600 & 1200 \leq x_R \end{cases} \end{aligned}$$

which is a concave piecewise linear function. Objective function $f(r_1, r_2, m_1, m_2)$ is sum of two concave functions, thus concave function. Our problem looks like:

$$\begin{aligned} \max \quad & f(r_1, r_2, m_1, m_2) \\ \text{s.t} \quad & \text{some constraints} \end{aligned}$$

We know that this problem can be reformulated as:

$$\begin{aligned} \min \quad & -f(r_1, r_2, m_1, m_2) \\ \text{s.t} \quad & \text{some constraints} \end{aligned}$$

where objective is now convex function $-f(r_1, r_2, m_1, m_2) = -f_M(m_1, m_2) - f_R(x_R)$. Function $-f_R(x_R)$ is convex piecewise function and hence can be written as maximum of its linear pieces:

$$-f(x_R) = \max\{50 - 3.05x_R, -3x_R, -2.5x_R - 600\}$$

Finally objective function (of our minimization problem) can be written as

$$-f(r_1, r_2, m_1, m_2) = \max\{50 - 3.05(r_1 + r_2), -3(r_1 + r_2), -2.5(r_1 + r_2) - 600\} - 2.2m_1 - 3.1m_2$$

Constraints:

- Yield of mangoes type 1: $r_1 + m_1 = 1000$
- Yield of mangoes of type 2: $r_2 + m_2 = 700$

- At least 60% of the total amount of mangoes sold to the retail company must be of type 2: $r_2 \geq 0.6(r_1 + r_2)$
- All mango quantities are nonnegative: $r_1, r_2, m_1, m_2 \geq 0$.

By putting it all together we get:

$$\begin{aligned}
 \min \quad & \max\{50 - 3.05(r_1 + r_2), -3(r_1 + r_2), -2.5(r_1 + r_2) - 600\} - 2.2m_1 - 3.1m_2 \\
 \text{s.t.} \quad & r_1 + r_2 = 1000 \\
 & m_1 + m_2 = 700 \\
 & r_2 \geq 0.6(r_1 + r_2) \\
 & r_1, r_2, m_1, m_2 \geq 0
 \end{aligned}$$

which we can reformulate as LP:

$$\begin{aligned}
 \min \quad & w - 2.2m_1 - 3.1m_2 \\
 \text{s.t.} \quad & r_1 + r_2 = 1000 \\
 & m_1 + m_2 = 700 \\
 & r_2 \geq 0.6(r_1 + r_2) \\
 & 50 - 3.05(r_1 + r_2) \leq w \\
 & -3(r_1 + r_2) \leq w \\
 & -2.5(r_1 + r_2) - 600 \leq w \\
 & r_1, r_2, m_1, m_2 \geq 0
 \end{aligned}$$

Note: There are multiple correct ways to formulate LP. Problem only asked for linear formulation, you were not required to implement it in cvxpy and provide solution. **Any correct linear formulation should be accepted.**

If you want to verify correctness of your formulation, you can implement it in cvxpy. You should obtain following solution:

Optimal net revenue is \$4,673.33.

Type 1 mangoes sold to a retail company: 466.66

Type 2 mangoes sold to a retail company: 700

Type 1 mangoes sold in a local market: 533.33

Type 2 mangoes sold in a local market: 0